

Assignment 1

CCS3308-Virtualization and Containers

Due Date: 15th August 2025

The Task

Deploy a Docker-based web application with at least two services and one persistent volume. Each service must:

- Listen on its own port.
- Run in a separate Docker container.
- Communicate with other required services.
- At least one service must maintain state using a persistent volume.

You may use existing Docker Hub images (e.g., for databases or web servers) or build custom images. Configure services using configuration files, startup arguments, or environment variables.

Files to Submit

Upload the following to your public GitHub repository:

1. **prepare-app.sh**: Bash script with Docker commands to:
 - Build images (if custom).
 - Create configurations, networks, and named volumes.
2. **start-app.sh**: Bash script to:
 - Start and configure all service containers.
 - Ensure containers restart on failure.
 - Display a message indicating the port for accessing the application via a web browser.
3. **stop-app.sh**: Bash script to:
 - Stop all services without resetting state (persistent data preserved).
 - Allow restarting without resetting to default state, even if configurations change.
4. **remove-app.sh**: Bash script to:
 - Remove all resources created by prepare-app.sh (images, containers, networks, volumes).
5. **docker-compose.yaml** (optional): If using Docker Compose or Swarm mode, include the application configuration.
6. **README.md**: Documentation file (see below for details).

Documentation (README.md)

Include the following in README.md:

- **Deployment Requirements:** Detailed list of required software (e.g., Docker, Docker Compose).
- **Application Description:** What the application does.
- **Network and Volume Details:** Description of virtual networks and named volumes used.
- **Container Configuration:** Details of how containers are configured.
- **Container List:** List of containers and their roles.
- **Instructions:**
 - How to prepare, run, pause, and delete the application.
 - How to access the application via a web browser.
- **Example Workflow:** Demonstrate application usage, e.g.:

Create application resources

`./prepare-app.sh`

Preparing app ...

Run the application

`./start-app.sh`

Running app ...

The app is available at `http://localhost:5000`

Open a web browser and interact with the application

Pause the application

`./stop-app.sh`

Stopping app ...

Delete all application resources

`./remove-app.sh`

Removed app.

Submission

- Create a public GitHub repository named <your_registration_number>.
- Upload all files to the repository.
- Submit the repository link via the LMS.
- Ensure the repository is accessible for evaluation.

Evaluation Criteria

- **Usability:** Do scripts execute without errors?
- **Completeness:** Do files meet all assignment requirements?
- **Functionality:** Does the web application work seamlessly?
- **Documentation:** Is the README complete and accurate?
- **Clarity:** Are scripts and documentation clear and readable?
- **Originality:** Is your personal contribution evident? Cite any external sources if inspired by existing solutions.
